

COMP6006 Final Assignment (Part B)

Critical Analysis of Real-Time Web Application Design

Total Marks: 10

Overview

In this assessment, students will critically analyse the design and implementation decisions made while developing the **Conference Connect** platform in Part A.

Students are expected to demonstrate deeper technical understanding, critical evaluation, and analysis of architectural decisions related to SPA development, authentication, real-time communication, and frontend state management.

Responses should move beyond simple description and include justification, evaluation, trade-offs, and discussion of limitations where appropriate.

Questions

Question 1 — Analysis of SPA Architecture and State Management (3 Marks)

Critically analyse the frontend architecture used in your application.

Your discussion should include:

- component structure and organisation
- state management approach
- routing/navigation design
- advantages and limitations of your implementation

You should also discuss how your chosen design affected maintainability, scalability, or user experience.

Question 2 — Analysis of Real-Time Communication Design (3 Marks)

Critically evaluate your Socket.io implementation.

Your discussion should include:

- how real-time communication was handled
- room/event design decisions
- participant synchronization
- challenges associated with real-time systems

- limitations or potential scalability concerns

You should also discuss how your implementation could be improved for larger-scale deployments.

Question 3 — Authentication and Security Considerations (2 Marks)

Analyse the authentication and route protection mechanisms used in your application.

Your response should include discussion of:

- session/authentication handling
- protected route implementation
- potential security risks
- limitations of the current design
- improvements that could strengthen security

Question 4 — Critical Reflection and Future Improvements (2 Marks)

Critically reflect on your overall implementation.

Discuss:

- the most technically challenging aspect of the assignment
- trade-offs made during development
- features or architectural improvements you would introduce in a production-ready system

Submission Requirements

- Submit as a PDF document
- Recommended length: approximately 1200–1800 words
- Academic writing style is expected
- Use diagrams, screenshots, or code snippets where appropriate to support analysis